

Moving point charge

Paul Haskins

March 12, 2025

Rel Lamor eq

$$P(t) = \frac{2}{3} \frac{q^2}{c} [\gamma^6 (\beta^{\cdot 2} - (\beta \times \beta^{\cdot})^2)]_{ret} \quad (1)$$

non-rel limit ie $\beta \ll 1$

$$P(t) \approx \frac{2}{3} \frac{q^2}{c} (\beta^{\cdot 2})_{ret} + O(\beta^2) \quad (2)$$

$$(\gamma^2)^2 = \frac{1}{(1 - \beta^2)^2} \approx 1 + 2\beta^2 + \dots \quad (3)$$

Circular motion

$|\vec{v}|w\rho = \text{const}$, ρ = radius.

$$\vec{r}(t) = \rho (\cos \phi(t) \sin \phi(t), 0)^T \quad (4)$$

$$\vec{v}(t) = \dot{\vec{r}}(t) = \rho \dot{\phi}(t) (-\sin \phi(t) \sin \phi(t), 0)^T \quad (5)$$

$$\ddot{\vec{r}}(t) = \rho \omega^2(t) (\cos \phi(t) \sin \phi(t), 0)^T = -\omega^2 \vec{r}(t) \quad (6)$$

with $\omega = \dot{\phi}$. Now we can look at the power

$$P = \frac{2}{3} \frac{q^2}{c} \gamma^6 (\beta^{\cdot 2} - (\beta \times \beta^{\cdot})^2) \quad (7)$$

if we have $\beta \perp \beta^{\cdot}$

$$= \frac{2}{3} \frac{q^2}{c} \gamma^6 (\beta^{\cdot 2} - \beta^2 \beta^{\cdot 2}) = \frac{2}{3} \frac{q^2}{c} \gamma^4 \beta^{\cdot 2} \quad (8)$$

This lead to instability of atoms in the classical understanding as electrons lose energy through radiation and spiral into the nucleus.

0.0.1 (Collider Experiments e.g LHC)

for the case of LHC $2\pi\rho \approx 27$ km. Proton on proton action ;), Pb-Pb collisions.

LEP electron on positron action

Example

Electron $E = 100$ GeV

$$\vec{\beta} = \frac{c\vec{p}}{E} = \frac{\sqrt{E^2 - m_e^2 c^4}}{E} = \sqrt{1 - \left(\frac{m_e c^2}{E}\right)^2} \approx \quad (9)$$

$$\frac{m_e c^2}{E} = \frac{0.5 \text{ MeV}}{100 \times 10^3 \text{ MeV}} = 0.5 \times 10^{-5} \implies \beta \approx 1 - 0.13 \times 10^{-10} \quad (10)$$

$$\gamma = \frac{E}{m_e c^2} \approx 2 \times 10^5 \implies \gamma^4 = 1.6 \times 10^{21} \quad (11)$$

Energy loss for electron $\beta \approx 1$

$$\Delta E \approx \frac{4\pi}{3} \left(\frac{q^2}{m_e c^2} \right) m_e c^2 \frac{\gamma^4}{\rho} \quad (12)$$

where the thing in brackets is 2×10^{-15} electron radius. And the $m_e c^2$ is 0.5 MeV

In the case of the LHC.

$$\rho \approx 5000m, \Delta E \approx 6 \times 10^{-15} \frac{\gamma^4}{(\frac{\rho}{m}) \text{ MeV}} = 2 \times 10^{-15+21-3-3} \text{ GeV} = 2 \text{ GeV}.$$

To keep ΔE small while increasing the energy on needs to increase $\rho \propto \gamma^4 \propto E^4$.

Example to double the energy $\rho \rightarrow 2^4 * \rho$.

Since $\gamma = \frac{E}{m c^2}$ particles with larger mass radiate less at the same energy.

$$\frac{\Delta E_{\text{elec}}}{\Delta E_{\text{protons}}} = \left(\frac{\gamma_{el}}{\gamma_{el}} \right)^2 \propto (2000)^4 = 16 \times 10^{12} = 1.6 \times 10^{13} \quad (13)$$